

Assignment # 2

Page Replacement Algorithm Simulator

Due Date: December 15, 2025

Details

Students are asked to write a program that simulates how an OS replaces pages in main memory when a new page is requested, and the memory is full.

The program should:

1. Take user input for:
 - Number of frames (memory slots available)
 - A sequence of page requests (e.g., 7, 0, 1, 2, 0, 3, 0, 4, ...)
2. Implement the following algorithms:
 - FIFO (First-In-First-Out)
 - OPT (Optimal)
 - LRU (Least Recently Used)
3. Show step-by-step how pages enter and leave memory.
4. Output:
 - Page table after each request
 - Total number of page faults
5. Analysis:
 - Compare FIFO, LRU, and Optimal, and show which gives the fewest page faults with a visualization

Deliverables

Submit the following:

1. `source` — Actual code in Python/C/Java
2. Screenshot of the execution of the program capture
3. A `report.pdf` file containing all the steps you have performed and an explanation of the code.